

Attractor networks

The simplest network model one can formulate is completely linear,

$$\tau_s \frac{du_i}{dt} + u_i = \sum_{j=1}^N T_{ij} v_j + h_i$$

where u_i is synaptic activity, and τ_s is a synaptic time constant. The synaptic activity in neuron i due to neuron j depends on the presynaptic firing rate v_j and the ‘weight’ of the synaptic connection T_{ij} . Assuming that firing rate is a linear function of synaptic activity,

$$v_i = g_i u_i + v_i^0$$

an equation for dv_i/dt can be derived whose steady state is given by solutions to

$$v_i = \sum_{j=1}^N W_{ij} v_j + f_i \quad (1)$$

where $W_{ij} = g_i T_{ij}$ and $f_i = g_i h_i v_i^0$.

In matrix form, Eq. 1 can be written as $\mathbf{v} = \mathbf{W}\mathbf{v} + \mathbf{f}$, which is equivalent to

$$(\mathbf{I} - \mathbf{W})\mathbf{v} = \mathbf{f}, \quad (2)$$

where $\mathbf{I}$ is an $N \times N$ identity matrix.

Steady states

Eq. 2 is a linear algebra problem where the vector $\mathbf{f}$ and matrix $\mathbf{I} - \mathbf{W}$ are given, while the vector $\mathbf{v}$ is unknown. The associated homogeneous system is

$$(\mathbf{I} - \mathbf{W})\mathbf{v} = \mathbf{0}. \quad (3)$$

From linear algebra we know that if $\mathbf{I} - \mathbf{W}$ is full rank (i.e., has a non-zero determinant), the only solution to Eq. 3 is $\mathbf{v} = \mathbf{0}$. Furthermore, there is exactly one solution to Eq. 2 given by

$$\mathbf{v} = (\mathbf{I} - \mathbf{W})^{-1} \mathbf{f}. \quad (4)$$

This is the “generic case,” meaning that for most choices of $\mathbf{W}$, the matrix $\mathbf{I} - \mathbf{W}$ will have an inverse, there is precisely one steady-state of Eq. 2, and this steady state is given by Eq. 4. The steady state $\mathbf{v}$ can be thought of as *point* in N -dimensional Euclidean space.

If $\mathbf{I} - \mathbf{W}$ is not full rank, i.e., $\det(\mathbf{I} - \mathbf{W}) = 0$, the linear algebraic system given by Eq. 2 is under-determined. In this case, the inverse $(\mathbf{I} - \mathbf{W})^{-1}$ is not defined and there are either no solutions to Eq. 1, or an infinite number of solutions. If $\mathbf{f}$ is not in the range of $\mathbf{I} - \mathbf{W}$, Eq. 2 has no solution. If $\mathbf{f}$ is in the range of $\mathbf{I} - \mathbf{W}$, every solution of Eq. 2 will have the form

$$\mathbf{v} = \mathbf{v}_p + \mathbf{v}_h \quad (5)$$

where $\mathbf{v}_p$ is a particular solution of Eq. 2, and $\mathbf{v}_h$ is a solution of the homogeneous equation Eq. 3.

If the rank of $\mathbf{I} - \mathbf{W}$ is $N - 1$, then the nullspace of $\mathbf{I} - \mathbf{W}$ is 1-dimensional. In that case, every solution of the homogeneous equation is a scalar multiple of a single vector. Denoting this vector as $\mathbf{v}_h^*$, the general solution of Eq. 2 is $\mathbf{v} = \mathbf{v}_p + c\mathbf{v}_h^*$ for arbitrary constant c . This is a *line* of steady-states in N -dimensional Euclidean space.